

# Renewable Energy Sources Microgrid Design for Rural Area in South Africa

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**Abstract-** Approximately 1.4 billion people around the world lack access to electricity, of which 85% are rural dwellers, mostly living in Sub-Saharan Africa. In South Africa, 55% of rural dwellers lack access to electricity. The Umhlabuyalingana Local Municipality is the least electrified municipality in the country with an electrification rate of 20%. It is therefore taken as a case study, investigating the implementation of a Renewable Energy Sources (RES) microgrid compared to grid extension. HOMER software was the tool used to carry out the simulation, optimization and sensitivity analyses in this research. It was discovered that a Photo Voltaic (PV) with Battery system is the optimal microgrid combination for the proposed microgrid yielding \$0.378/kWh cost of electricity, 0 kg/person CO<sub>2</sub> emission, 100% renewable penetration compared to \$0.328/kWh cost of grid electricity, 8.9 kg/person CO<sub>2</sub> emission from grid extension and 0% renewable penetration from the national grid. The microgrid had a standalone breakeven Electric Distance Limit of 34 km less than the 150 km transmission powerline required for uMhlabuyalingana. This microgrid design is proposed as a better solution to electricity access in unelectrified areas of Umhlabuyalingana local municipality than grid extension.

**Keywords-** Microgrid, electricity access, renewable energy sources.

## I. INTRODUCTION

The socio-economic development of a country depends greatly on its level of electricity access. There are about 1.4 billion people around the world that lack access to electricity, (85% being rural dwellers), with the majority of whom are living in Sub-Saharan Africa [1]. Universal electricity access to rural areas through grid extension has been impeded by her spatial heterogeneity, high costs of electricity generation, transmission, and distribution, low energy demand, very low level of industrialization; low return-on-investment for investors; high operation and maintenance cost of grid electricity; high CO<sub>2</sub> emissions and also voltage loss due to long distance transmission; among others.

The International Energy Agency (IEA) [2] reports that South Africa had an electrification rate of 75% in 2009, which means that 12.3 million of its population had no access to electricity. Census 2011 of South Africa showed that 84.7% of households in South Africa use electricity for lighting [3]. Also in [4], it was reported that 80% of urban dwellers and 45% of rural dwellers in South Africa have access to electricity, which is a similar situation with other

developing countries with most unelectrified areas being rural areas. In South Africa, Umhlabuyalingana Local Municipality is the least electrified municipality in the country with an electrification rate of 20% [3].

Renewable Energy Technology (RET) needs to be given more attention as the nation pursues her goal of universal electricity access. Such attention might include encouragement of local production of RET equipment and accessories. RET microgrids needs to spread much more across the country especially to rural areas where the comparative cost of standalone microgrid could be lesser than grid extension. Therefore, RET microgrid is presented here as a local solution to electricity access in other rural unelectrified areas of South Africa and most developing countries of the world.

In this paper, a comparison of electricity access through grid extension and a Renewable Energy Sources (RES) microgrid to uMhlabuyalingana Local Municipality is simulated to find the optimum techno-economic option in meeting the electricity access needs of the people in the area.

The paper is organized as follows: an introduction on uMhlabuyalingana Local Municipality is presented in Section II while a simulation for the proposed RES microgrid simulation is discussed in Section III. Sections IV, V and VI contain the microgrid load profile, optimization analysis and sensitivity analysis respectively. The results are analyzed in Section VII while the conclusion is in Section VIII.

## II. uMHLABUYALINGANA LOCAL MUNICIPALITY

uMhlabuyalingana Local Municipality is one of the five municipalities in uMkhanyakude District Municipality, situated in the North-Eastern part of Kwazulu-Natal Province of South Africa and lies on 27°1' S 32°44'E coordinates. The municipality is one of the poorest municipalities in the country; it encompasses iSimangaliso Wetland Park, a world heritage site. This municipality is made up of at least 99% black African people, most of whom are Zulu-speaking. The location of uMhlabuyalingana Local Municipality on the map of Kwazulu-Natal Province, South Africa, is shown in the shaded area of Figure 1.

The municipality is 99% rural and has a total land area of 3,964 km<sup>2</sup> [5]. The municipality has a population of 156,736

and 33,857 households with an average ratio of 6 persons per household.



Figure 1: Location of Umhlabuyalingana Local Municipality on Kwazulu-Natal Map

uMhlabuyalingana has an electrification backlog that is mainly attributed to the previous neglect of the area and the low density settlement patterns which renders electrification costs high. The area is sparsely settled and, in some instances, the terrain is very difficult. The limited spare capacity of the existing electricity network has also been a factor in this regard. The nodal areas, particularly Manguzi and Mbazwana, are the most affected by power shortages due to a relatively high demand for energy in these areas. There is a general perception that the quality of supply is unreliable and too intermittent to encourage investor confidence [5].

### III. RES MICROGRID SIMULATION FOR UMHLABUYALINGANA LOCAL MUNICIPALITY

This research, using renewable energy sources (RES) for a microgrid design, can contribute to reduction in emission of green house gases and preservation of our environment. Wind and solar energy are the RES components used in this microgrid design. Hourly wind and solar data for uMhlabuyalingana were not readily available therefore locations whose data were available were chosen that fell in the same solar and wind belts with uMhlabuyalingana. Therefore, time series (8,760 hours) wind data of Sutherland was obtained from the Wind Association of South Africa (WASA) [6] and simulated for uMhlabuyalingana. Also, time series (8,760 hours) solar data of Lephalale was obtained from Council for Scientific and Industrial Research (CSIR) [7] and simulated for uMhlabuyalingana. The simulated scaled annual average sunshine irradiation and wind speed for uMhlabuyalingana Local Municipality are 5900 kWh/m<sup>2</sup>/day and 7.06 m/s respectively. Average monthly radiation and wind speed for uMhlabuyalingana are shown in Figures 2 and 3.

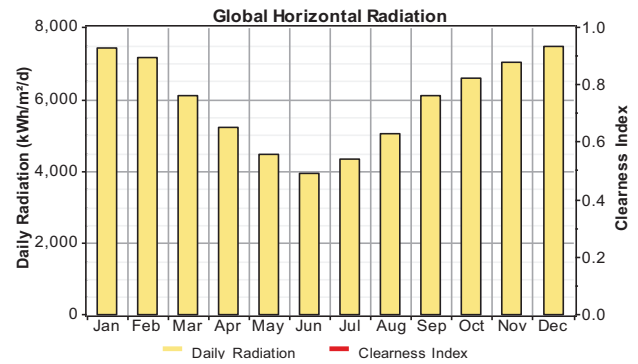


Figure 2: Monthly Daily Radiation for uMhlabuyalingana

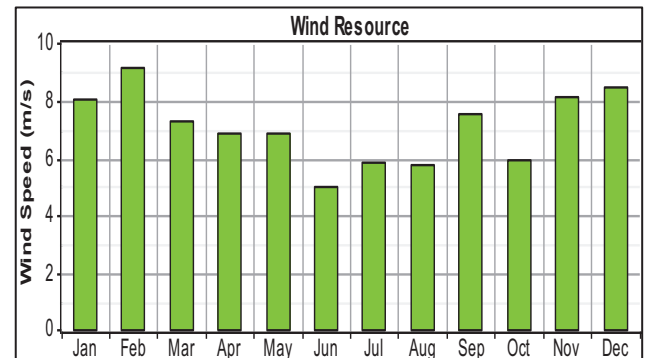


Figure 3: Monthly wind speed for uMhlabuyalingana

Due to the variations in the availability of wind and solar energy resources at different times of the day and year, a battery is included in the design for storage. Lastly, a diesel generating plant is also included in the microgrid design as a back-up in case there is a failure from the above three energy sources (solar, wind and battery), especially in the winter months when load is expected to be higher. A converter/inverter is also included for conversion of alternating current to direct current and vice versa.

Hybrid Optimization Model for Electric Renewable (HOMER) software was developed by National Renewable Energy Laboratory (NREL), U.S.A to assist in the design of micropower systems and to enhance the comparison of power generation technologies across a wide range of applications [8]. HOMER software is used for the simulation and optimization analyses in this design because it limits the input complexity, performs fast enough computation using a logic that is less detailed compared to several other time-series simulation models being used for micropower systems like Hybrid2, PV-DesignPro and PV\*SOL and even non-time series simulators like RETScreen [8]. HOMER is also able to display a ranked optimization result in an increasing order of total Net Present Cost (NPC), listing in a table the feasible design options starting with the one with the lowest NPC as the optimal system configuration. The proposed uMhlabuyalingana microgrid as simulated in HOMER is shown in Figure 4.

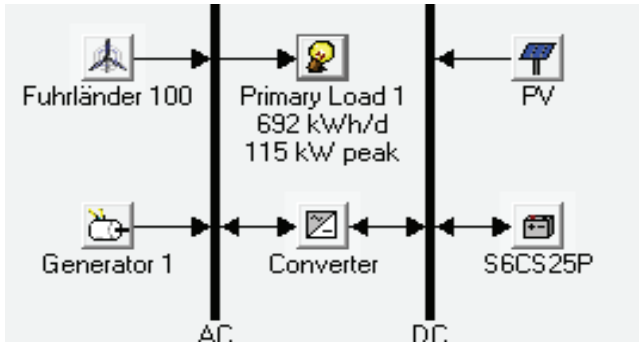


Figure 4: Schematic Representation of Proposed uMhlabuyalingana Microgrid

HOMER calculates the output power of the Photo Voltaic (PV) system,  $P_{PV}$  array using equation (1) [8]:

$$P_{PV} = f_{PV} Y_{PV} \frac{I_T}{I_S} \quad (1)$$

Where  $f_{PV}$  is the PV derating factor,  $Y_{PV}$  is the rated capacity of the PV array (kW),  $I_T$  is the global solar radiation (beam plus diffuse) incident on the surface of the PV array ( $\text{kW/m}^2$ ) and  $I_S$  is  $1 \text{ kW/m}^2$ , which is the standard amount of radiation used to rate the capacity of the PV array. The generator's fixed cost of energy,  $c_{gen, fixed}$  given as [8]:

$$c_{gen, fixed} = c_{om, gen} + \frac{c_{rep, gen}}{R_{gen}} + F_0 Y_{gen} c_{fuel, eff} \quad (2)$$

Where  $c_{om, gen}$  is the operation and maintenance (O&M) cost (\$/hr),  $c_{rep, gen}$  is the replacement cost (\$),  $R_{gen}$  is the generator lifetime (hrs),  $Y_{gen}$  is the capacity of the generator (kW) and  $c_{fuel, eff}$  is the effective price of fuel (\$/litre). The effective price of fuel includes the cost penalties, if any, associated with the emissions of pollutants from the generator. The marginal energy cost of the generator,  $c_{gen, mar}$  is given as [8]:

$$c_{gen, mar} = F_1 c_{fuel, eff} \quad (3)$$

Also, HOMER assumes that inverter and rectifier capacities are not surge capacities, but rather continuous capacities that the device can withstand for as long as necessary.

#### IV. uMHLABUYALINGANA MICROGRID LOAD PROFILE

Microgrid analysis of universal electricity access in uMhlabuyalingana requires time series data also for household load profile. Daily, seasonal and annual electrical load profiles were generated and simulated with Hybrid Optimization Model for Electric Renewable (HOMER). The total energy consumed,  $E_T$  by a household is given as:

$$E_T = \sum_{i=1}^m n P_i T_i = E_1 + E_2 + E_3 + \dots E_m \quad (4)$$

Where  $n$  is the quantity of the appliance  $i$  in use,  $P_i$  is the power rating of appliance  $i$  and  $T_i$  is the duration of appliance  $i$  usage.

TABLE I. HOUSEHOLD ELECTRICAL LOAD

| Load    | Rating<br>$P$ (kW) | Usage<br>$T$ (hrs) | Quantity/<br>household $n$ | Energy<br>$E = nPT$<br>(kWh/d) |
|---------|--------------------|--------------------|----------------------------|--------------------------------|
| Radio   | 0.015              | 2                  | 1                          | 0.030                          |
| TV      | 0.040              | 5                  | 1                          | 0.200                          |
| Fridge  | 0.150              | 24                 | 1                          | 3.600                          |
| Stove   | 1.000              | 2                  | 1                          | 2.000                          |
| Phone   | 0.010              | 2                  | 2                          | 0.040                          |
| Bulbs   | 0.020              | 6                  | 2                          | 0.240                          |
| Iron    | 1.000              | 1                  | 1                          | 1.000                          |
| Kettle  | 1.000              | 2                  | 1                          | 2.000                          |
| Fan*    | 0.080              | 3                  | 1                          | 0.240                          |
| Heater* | 1.000              | 11                 | 1                          | 11.000                         |

\*Indicates seasonal loads.

Only 3% of a total of 52% economically active population within the Municipality earns more than \$162 per month [5]. It was assumed that the same appliances are owned by all households and usage was similar for all households. The basic appliances that constitute the load are shown in Table I. The average annual daily load profile as simulated by HOMER for fifty (50) households is shown in Figure 5.

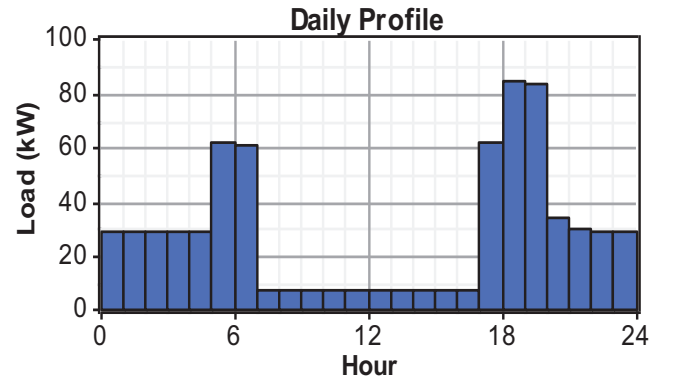


Figure 5: Average Daily Load Profile

#### V. uMHLABUYALINGANA MICROGRID OPTIMIZATION ANALYSIS

HOMER was used to simulate different system configurations to obtain the one that satisfies the specified technical constraints at the least life-cycle cost (LCC) and hence the least Net Present Cost (NPC) is obtained. HOMER calculates the total NPC using equation (5) [8]:

$$C_{NPC} = \frac{C_{ann, tot}}{CRF(i, R_{proj})} \quad (5)$$

Where  $C_{ann,tot}$  is the total annualized cost,  $i$  is the annual real interest rate (discount rate),  $R_{proj}$  is the project lifetime and  $CRF(i, R_{proj})$  is the capital recovery factor and is given by equation (7), where  $N$  is the number of years:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (6)$$

In the overall optimization result displayed, HOMER lists the feasible system configurations from the most cost-effective (least NPC) to the least cost-effective (highest NPC).

Table II is the tabular result of the summarized categorized optimization result of the microgrid. The most optimal feasible combination for the proposed microgrid is the highlighted first row of Table II, and it is the PV/Battery combination having the least total NPC and cost of electricity (COE) of \$0.378/kWh and a renewable penetration (or Renewable Fraction) of 1.00 since the optimal design comprised only of PV cells and a battery.

TABLE II. OPTIMIZATION RESULT FOR PROPOSED uMHLABUYALINGANA MICROGRID

| Microgrid Generation Mix | Total NPC (\$) | COE (\$/kWh) | Renewable Fraction |
|--------------------------|----------------|--------------|--------------------|
| PV/Battery               | 1,220,029      | 0.378        | 1.00               |
| PV/DG/Battery            | 1,240,485      | 0.394        | 1.00               |
| PV/Wind/DG/Battery       | 1,562,204      | 0.414        | 0.93               |
| PV/Wind/Battery          | 1,732,445      | 0.459        | 1.00               |
| Wind/DG/Battery          | 2,416,412      | 0.640        | 0.64               |
| PV/Wind/DG               | 4,555,533      | 1.207        | 0.51               |
| DG/Battery               | 5,391,039      | 1.428        | 0.00               |
| Wind/DG                  | 5,950,324      | 1.576        | 0.36               |
| PV/DG                    | 10,178,164     | 2.696        | 0.00               |
| DG                       | 17,591,758     | 4.661        | 0.00               |

## VI. uMHLABUYALINGANA SENSITIVITY ANALYSIS

The sensitivity analysis was done using HOMER to investigate the effect of uncertainties or variations in the five major variables, (namely, average wind speed, average solar radiation, electric load, fuel price and grid extension cost), would have on the choice of micropower design. For this analysis, minimum and maximum values were chosen for the five variables and the results are shown in Figs. 6 and 7 respectively with the COE superimposed on the plots.

## VII. DISCUSSION

The findings of this research are discussed under the following:

**A. CO<sub>2</sub> and Other Greenhouse Gases Emissions:** The Umhlabuyalingana microgrid design showed a renewable penetration of 1.00 for the optimal design, which implies that emissions from greenhouse gases such as CO<sub>2</sub>, Carbon

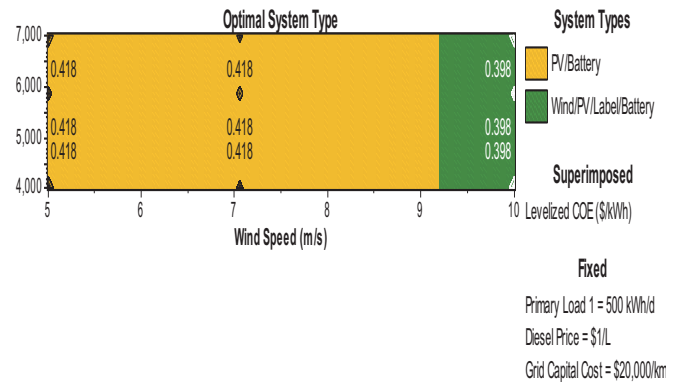


Figure 6: Optimal System Graph for Proposed uMhlabyalingana Proposed Microgrid (Minimum values)

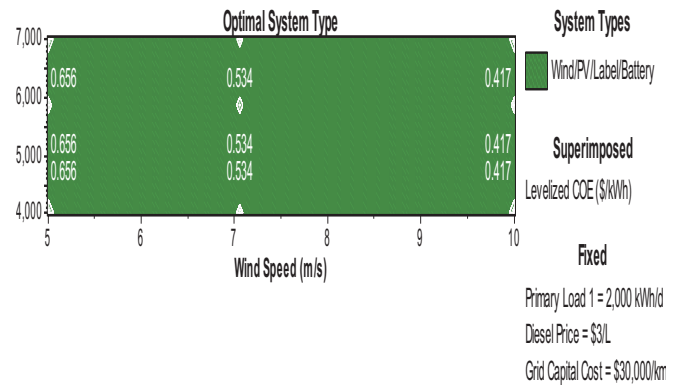


Figure 7: Optimal System Graph for Proposed uMhlabyalingana Proposed Microgrid (Maximum values)

monoxide, unburned hydrocarbons, Particulate matter, Sulphur dioxide and Nitrogen oxides are 0.00. Hence, this microgrid is highly safe for human health in Umhlabuyalingana, compared to the 8.9 kg/person CO<sub>2</sub> emissions and 0.99 kg/kWh from grid extension in South Africa in 2012 [9].

**B. Breakeven Distance for Grid Extension:** The breakeven distance or otherwise called the Electric Distance Limit (EDL) in this analysis was found to be 34 km as shown in Figure 8. This implies that it is only comparatively advantageous to connect uMhlabyalingana to the traditional grid provided the distance between the source substation and uMhlabyalingana is  $\leq 34$  km.

Since most of the unelectrified areas in uMhlabyalingana are hundreds of km from grid and at such distances, the cost of grid extension far exceeds the cost of the proposed standalone microgrid. The national electricity provider, Eskom in [10] proposed constructing a 132kV powerline in the rural area of the Jozini and uMhlabyalingana Municipalities as well as two associated 132/22kV substations in the uMhlabyalingana Municipality on the Makhathini Flats. The proposed transmission powerline will cover a distance of approximately 150 km and have a servitude width of 32m,



which will in turn be more expensive (higher total NPC) than the proposed ULM microgrid.

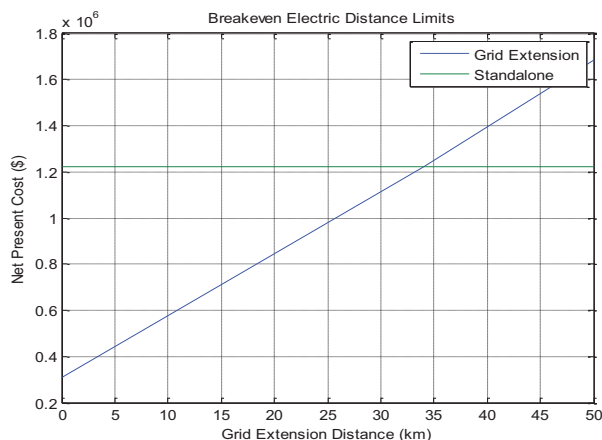


Figure 8 Graph Showing Electric Distance Limit (EDL) for Proposed uMhlabuyalingana Microgrid

Susanto [11] reported the cost of a single wire earth return (SWER) to be \$18,544/km. The average cost of grid extension in Sub-Saharan Africa has been estimated by [12] as \$20,000/km for an 11 kV line cost only. Adding other required electrification components increases the total estimate to \$25,000/km. The cost of extending the grid to mountainous areas is generally higher than regions with good terrains and Umhlabuyalingana having some areas with very difficult terrain may cost more.

## VIII. CONCLUSION

This study has shown that for an unelectrified locality like Umhlabuyalingana local municipality, a stand-alone microgrid is an optimum solution to bring electricity access to rural areas. The best option found out in this research is the PV/Battery microgrid for Umhlabuyalingana where the PV system would generate all the energy production. The cost of energy to the consumers at the least NPC was \$0.378/kWh compared to the cost of \$0.328/kWh if the area is to be connected to the grid. This difference of \$0.05/kWh COE is an advantage to a potential commercial provider of the microgrid as the Renewable Energy Feed-in-Tariff (REFIT) of the government would yield a comparative long-term advantageous COE less than grid COE and total NPC. Hence, this RES microgrid would be a more cost effective option. Also, its 100% renewable penetration means 0 kg/person CO<sub>2</sub> emission compared to the general average of 8.9 kg/person CO<sub>2</sub> emission in South Africa. The extension of the existing electrical grid to rural areas is not always the best option considering the estimated capital cost of grid extension of about \$25,000 per km, which can be higher for a difficult terrain like uMhlabuyalingana Local Municipality. With the implementation of the microgrid, transmission losses associated with grid extension would also be eliminated.

The population of uMhlabuyalingana is very young with 44% younger than 15, and 77% are younger than 35 years

old. Hence, electricity access will lead to youth empowerment, increase education levels and opportunities and positively influence the municipality towards a better tomorrow.

The long-term feasibility of off-grid depends on other factors such as availability of repair facilities, availability of maintenance personnel, communal influences etc, which were not considered in this report.

It is hereby proposed to always carry out techno-economic analysis such as this for any rural unelectrified area in the world, especially in developing countries, before extending grid there so that the cost is optimized both for the utility provider and consumers.

Further work to improve the RES microgrid include research into how the high cost of battery can be reduced for the microgrid, a real time control over demand and generation and also better energy policy proposals for decision makers in the energy industry of South Africa and other developing nations.

## IX. ACKNOWLEDGEMENT

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